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Module 2 Assignment: Law Enforcement Equipment

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Lesson A: The Bullet-Resistant Vest

Use the following information to answer the next question.

Various Molecular Traits

1. Natural polymer
2. Synthetic polymer
3. Radial composition
4. Square composition
5. Contains hydrogen bonds
6. Contains only single bonds
7. Fibres oriented in a circular pattern
8. Fibres oriented in a parallel arrangement

1. Four molecular traits of Kevlar, as shown above, include _____ , _____ , _____ , _____ , and _____. (2 marks)

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Use the following information to answer the next question.

Various Descriptions Related to Bullet-Resistant Vests

1. Invented unintentionally by a chemist in the US
2. Durable and lightweight; ideal for police officers
3. Can absorb energy delivered by rifle rounds or knives

2. Match each numbered description with the correct material or object as given below: (3 marks)

Material/Object	Description
Soft body armour	
Kevlar	
Hard body armour	

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3. Identify two advantages and two disadvantages of Kevlar in bullet-resistant vests. (4 marks)

4. Explain what a trauma plate is and why it is inserted into a bullet-resistant vest. (2 marks)

5. Identify two factors that can cause the breakdown of Kevlar over time, and describe how bullet-resistant vests have been designed to prevent this (3 marks)

6. Why was the testing of the first bullet-resistant vest designed for police officers controversial? (2 marks)

Lesson B: The Conducted Energy Device (CED)

Use the following information to answer the next question.

Steps Occurring inside a Conducted Energy Device (CED):

1. **The trigger of the CED is activated.**
2. **Electric current flows into a step-up transformer.**
3. **The voltage of the power supply increases.**
4. **The oscillator converts energy from the batteries into a stable electrical resonant circuit.**

7. The correct sequence of the steps shown above occurring in a functioning CED are _____, _____, and _____. (2 marks)

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Use the following information to answer the next question.

Various Descriptions of Electroshock Devices:

- 1. Wireless projectile that can be fired from a 12-gauge shotgun**
- 2. Handled electroshock weapon used by police to subdue subjects**
- 3. Stand-alone device that observes, warns, incapacitates, and retains intruders**

8. Match each electroshock device description numbered above with the appropriate device in the chart below: (3 marks)

Device	Description
CED	
XREP	
TRAD	

9. Describe how the symptoms of being contacted with the CED for one-second differ from the symptoms of a five-second CED contact. (2 marks)

10. Identify three advantages to the use of CEDs by police officers. (3 marks)

11. Identify three factors that affect the electrical current experienced by a subject hit by a CED (3 marks)

12. Describe two situations that warrant the use of a CED by a police officer. (2 marks)

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Lesson C. Pepper Spray and Tear Gas

Use the following list to answer the next questions.

Various Traits of OC Pepper Spray and CS Tear Gas

1. **Natural compound**
2. **Synthetic compound**
3. **Irritates the stomach**
4. **Irritates the eyes and nose**
5. **Soluble in fats and oils**
6. **Soluble in an alkaline solution**
7. **Used in riot control**
8. **Used upon unruly individuals**

13. a. Four unique traits of OC in pepper spray include ____, ____, ____ and _____. (2 marks)

b. Four unique traits of CS in tear gas include ____, ____, ____, and _____. (2 marks)

Note: The same trait can be used in both part a and b.

14. Why is a pepper spray with a lower concentration of OC often more effective than a pepper spray with a higher OC concentration? (2 marks)

15. Explain why tear gas is often dispersed using grenades. (2 marks)

16. Explain why OC pepper spray and CS tear gas are both considered “less-than-lethal” weapons (2 marks)

17. Identify one similarity and one difference between pepper spray and tear gas (2 marks)

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Lesson D. Case Study Involving Law Enforcement Safety Equipment

Use the following list to answer the next question:

Possible Descriptions of the North Hollywood Shootout

- 1. A bank security guard called police**
- 2. A witness outside the bank contacted the police**
- 3. Both suspects were shot and killed**
- 4. Both suspects survived and were apprehended**
- 5. The case involved 32 police officers**
- 6. The case involved 370 police officers**
- 7. Each suspect was wearing soft body armour**
- 8. Only one of the suspects was wearing body armour**

18. The four descriptions numbered above that describe the North Hollywood Shootout are ____, ____, ____, and _____. (2 marks)

19. In the North Hollywood Shootout, explain why police did not use pepper spray, tear gas, or police canines to subdue the two suspects in this case (3 marks)

Use the following list to answer the next question:

Possible Events that occurred in the Party Out of Control Incident

- 1. The hall is set on fire**
- 2. A fist fight begins**
- 3. The hall owner calls the police**
- 4. The crowd develops a mob mentality**
- 5. Several hall patrons make threatening gestures and insults**
- 6. Several officers discharge pepper spray into the crowd**
- 7. A police supervisor uses a fogger to disperse tear gas into the crowd**
- 8. A police supervisor uses a fogger to disperse pepper spray in the crowd**

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20. The correct sequence of events that occurred in the Party Out of Control Incident is ____, ____, ____, and _____. (2 marks)

Use the following list to answer the next question:

Possible Events that occurred in the Carl Rowland Incident.

- 1. Carl Rowland fled from police to Florida**
- 2. Carl Rowland ran from authorities in Georgia**
- 3. Carl Rowland's wife was sexually assaulted**
- 4. Carl Rowland's girlfriend was strangled to death**
- 5. Rowland was pepper sprayed and taken into custody**
- 6. Rowland is tasered and lowered from the crane**
- 7. Rowland pulled a knife on a group of construction workers**
- 8. Rowland took a crane operator hostage at a construction site**

21. The correct sequence of events numbered above that occurred in the Carl Rowland Incident ____, ____, ____, and _____. (2 marks)

22. In the Carl Rowland Incident, why did the police lure the suspect before apprehending him (3 marks)

23. Provide 3 reasons why the handling of the Party Out of Control Incident was an effective use of non-lethal force. (3 marks)